Fidelity total anticoherence is nondecreasing under covariant noise at order two

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Abstract

We prove the fidelity part of Conjecture 1 of *Total, quantum, and classical measures of anticoherence for mixed spin states*, at $t = 2$, for every $N \geq 3$. The notation and the squared Uhlmann–Jozsa fidelity normalization are those of `resub_v0.tex`. The proof is based on `su2_t2_full_direct_proof.pdf`. It treats arbitrary covariant channels directly, derives their two relevant multipliers without $6j$ symbols, and ends with an elementary three-eigenvalue argument.

1 Statement and conventions

As in the manuscript, $\mathcal{H}_j$ is the spin-j Hilbert space, $N = 2j$, and ρ_S is the state corresponding to $\rho \in \mathcal{D}(\mathcal{H}_j)$ under the identification $\mathcal{H}_j \simeq \mathrm{Sym}^N(\mathbb{C}^2)$. Put

$$\rho_t = \mathrm{Tr}_{N-t}(\rho_S), \qquad \rho_0^{(t)} = \frac{\mathbb{1}_{t+1}}{t+1}, \qquad F(\rho_t, \rho_0^{(t)}) = \frac{(\mathrm{Tr}\,\sqrt{\rho_t})^2}{t+1}.$$

Thus the fidelity total candidate of the manuscript is

$$\mathcal{A}_t^{T,F}(\rho) = \frac{(\mathrm{Tr}\,\sqrt{\rho_t})^2 - 1}{t}. \tag{1}$$

All determinants and traces of two-qubit reductions below are taken on the three-dimensional symmetric subspace, including any zero eigenvalues.

Theorem 1 (Conjecture 1, fidelity candidate, $t = 2$)**.** *Let $N \geq 3$, $j = N/2$, and let Φ be a completely positive, trace-preserving map on $\mathcal{B}(\mathcal{H}_j)$ satisfying*

$$\Phi(U\rho U^\dagger) = U\Phi(\rho)U^\dagger \quad \textit{for every spin-}j\textit{ rotation } U \textit{ and every state } \rho.$$

Then, for every $\rho \in \mathcal{D}(\mathcal{H}_j)$,

$$\boxed{\mathcal{A}_2^{T,F}(\Phi(\rho)) \geq \mathcal{A}_2^{T,F}(\rho).} \tag{2}$$

The condition $N \geq 3$ is exactly the proper-order condition $2 \leq t < N$ at $t = 2$. No restriction to pure inputs or to random rotations is made. The theorem does not address the trace-distance candidate.

2 The two channel multipliers

Use the Hilbert–Schmidt orthonormal irreducible tensor operators T_{LM} of the manuscript. The conjugation representation on $\mathcal{B}(\mathcal{H}_j)$ contains one copy of each rank $L = 0, \ldots, N$. Schur's lemma and Hermiticity preservation therefore give real numbers f_L such that

$$\Phi(T_{LM}) = f_L T_{LM}.$$

Covariance implies that $\Phi(\mathbb{1})$ is a scalar multiple of $\mathbb{1}$; trace preservation fixes that scalar to one. In particular $f_0 = 1$. Write

$$f = f_1, \qquad g = f_2, \qquad S = N(N+2), \qquad D = S(S-3) > 0.$$

The symbol S here is a scalar, distinct from the subscript in ρ_S.

Lemma 1 (Multiplier bounds)**.** *Every channel in Theorem 1 satisfies*

$$-\frac{3}{5} \le g \le 1, \qquad 3|f| \le 2 + g, \tag{3}$$

$$3Sf^2 - 6f - S \le 2(S-3)g. \tag{4}$$

Consequently $|f| \le 1$ *and* $|g| \le 1$*; if* $g < -1/2$*, then* $|f| < 1/3$*.*

Proof. We first derive formulas that include all covariant channels. Set $d = N + 1$ and $c = j(j+1) = S/4$. Let C_Φ be the unnormalized Choi matrix, with convention

$$\mathrm{Tr}[C_\Phi(B \otimes C^{\mathsf{T}})] = \mathrm{Tr}[B\Phi(C)]. \tag{5}$$

The unitary $V|j,m\rangle = (-1)^{j-m}|j,-m\rangle$ obeys $VJ_a^{\mathsf{T}}V^\dagger = -J_a$. Consequently $\widetilde{C}_\Phi = (\mathbb{1} \otimes V)C_\Phi(\mathbb{1} \otimes V^\dagger)$ commutes with $U \otimes U$. In $\mathcal{H}_j \otimes \mathcal{H}_j$, weight $m \in \{-N, \dots, N\}$ has multiplicity $N + 1 - |m|$. Subtracting consecutive nonnegative-weight multiplicities gives one copy of each spin $K = 0, \dots, N$. Thus the decomposition $\mathcal{H}_j \otimes \mathcal{H}_j = \bigoplus_{K=0}^N \mathcal{H}_K$ is multiplicity free. Positivity of the Choi matrix and $\mathrm{Tr}\, C_\Phi = d$ imply

$$\widetilde{C}_\Phi = d\sum_{K=0}^{N} p_K \frac{P_K}{2K+1}, \qquad p_K \ge 0, \qquad \sum_{K=0}^{N} p_K = 1, \tag{6}$$

where P_K projects onto total spin K. We use $\mathbb{E}$ for this finite weighted average and put $\chi_K = K(K+1)$.

For completeness the required Choi moments can be computed using only angular-momentum commutators. Define

$$X = \sum_{a=1}^{3} J_a \otimes J_a, \qquad Q_{ab} = \frac{J_aJ_b + J_bJ_a}{2} - \frac{c}{3}\delta_{ab}\mathbb{1}.$$

The total-spin Casimir gives $XP_K = (\chi_K/2 - c)P_K$. The J_a span rank one, whereas the symmetric traceless Q_{ab} span rank two. Moreover

$$\sum_{a,b} Q_{ab} \otimes Q_{ab} = X^2 + \frac{X}{2} - \frac{c^2}{3}\mathbb{1}, \qquad \sum_{a,b} Q_{ab}^2 = \frac{c(4c-3)}{6}\mathbb{1}. \tag{7}$$

Here the first identity follows by expanding the symmetrized products and using $[J_a, J_b] = i\sum_k \epsilon_{abk}J_k$. For the second, expansion of $\sum_b [J_b, [J_b, J_a]] = 2J_a$ gives $\sum_b J_bJ_aJ_b = (c-1)J_a$. Thus, writing $B_{ab} = (J_aJ_b + J_bJ_a)/2$,

$$\sum_{a,b} B_{ab}^2 = \frac{1}{2}\sum_{a,b}(J_aJ_bJ_aJ_b + J_aJ_bJ_bJ_a) = \left(c^2 - \frac{c}{2}\right)\mathbb{1},$$

which proves the second identity in (7).

Time reversal changes the sign of J_a^{T} and fixes Q_{ab}^{T}. Applying (5) therefore yields

$$-dcf = \mathrm{Tr}(\widetilde{C}_\Phi X) = d\mathbb{E}(\chi/2 - c),$$

and

$$\frac{dc(4c-3)}{6}g = d\mathbb{E}\left[(\chi/2-c)^2 + \frac{\chi/2-c}{2} - \frac{c^2}{3}\right].$$

Equivalently,

$$f = \mathbb{E}f_K, \quad g = \mathbb{E}g_K, \qquad f_K = 1 - \frac{2\chi_K}{S}, \quad g_K = 1 + \frac{6\chi_K(\chi_K - S + 1)}{D}. \tag{8}$$

Since $0 \le \chi_K \le N(N+1) < S-1$, we have $g_K \le 1$. For $N = 3$, the four values of g_K are $1, 1/5, -3/5, 1/5$. For $N \ge 4$, $S \ge 24$, completing the square gives

$$g_K \ge 1 - \frac{3(S-1)^2}{2S(S-3)} \ge -\frac{3}{5},$$

where the last inequality follows from

$$16S(S-3) - 15(S-1)^2 = (S-24)(S+6) + 129 > 0.$$

The two linear bounds follow on averaging the identities

$$2 + g_K - 3f_K = \frac{6\chi_K(\chi_K - 2)}{D} \ge 0, \qquad 2 + g_K + 3f_K = \frac{6[(\chi_K - S + 2)^2 + S - 4]}{D} > 0.$$

The first inequality uses the discrete fact that $\chi_K = 0$ or $\chi_K \ge 2$. Finally, direct substitution into (8) gives

$$2(S-3)g - (3Sf^2 - 6f - S) = \frac{12}{S}\mathbb{E}[(\chi - \mathbb{E}\chi)^2] \ge 0. \tag{9}$$

If $g < -1/2$, (4) implies $Sf^2 - 2f - 1 < 0$. Since $S \ge 15$, this forces $-1/3 < f < 1/3$: otherwise $Sf^2 - 2f - 1 \ge 15f^2 - 2f - 1 = (3f-1)(5f+1) \ge 0$. □

3 Physical two-qubit reductions

Write the symmetric two-qubit reduction in its Pauli form

$$\rho_2 = \frac{1}{4}\left[\mathbb{1}\otimes\mathbb{1} + \sum_{a=1}^{3} r_a(\sigma_a\otimes\mathbb{1} + \mathbb{1}\otimes\sigma_a) + \sum_{a,b=1}^{3} T_{ab}\sigma_a\otimes\sigma_b\right]. \tag{10}$$

Here $\mathbf{r} \in \mathbb{R}^3$, $T = T^{\mathsf{T}} \in M_3(\mathbb{R})$, and $\operatorname{Tr} T = 1$, since $\sum_a \sigma_a \otimes \sigma_a = 2\mathsf{F} - \mathbb{1}$ and the swap F is the identity on the symmetric subspace. In the orthonormal Cartesian spin-one basis

$$|x\rangle = \frac{|11\rangle - |00\rangle}{\sqrt{2}}, \quad |y\rangle = \frac{i(|11\rangle + |00\rangle)}{\sqrt{2}}, \quad |z\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}},$$

put $A = \mathbb{1}_3 - T$ and let $[\mathbf{r}]_\times \mathbf{v} = \mathbf{r} \times \mathbf{v}$. Substitution in (10) gives

$$2\rho_2 = A + i[\mathbf{r}]_\times, \qquad 8\det\rho_2 = \det A - \mathbf{r}^{\mathsf{T}} A\mathbf{r}. \tag{11}$$

To check the determinant identity, diagonalize A by a proper real rotation. The matrix then has diagonal a, b, c and off-diagonal entries $-ir_3, ir_2, -ir_1$ and their conjugates; its determinant is $abc - ar_1^2 - br_2^2 - cr_3^2$.

There is a useful bound specific to reductions of symmetric N-qubit states. With $J_a = \frac{1}{2}\sum_{n=1}^{N}\sigma_a^{(n)}$, permutation symmetry gives

$$\langle J_a\rangle = \frac{N}{2}r_a, \qquad \left\langle \frac{J_aJ_b + J_bJ_a}{2}\right\rangle = \frac{N}{4}\delta_{ab} + \frac{N(N-1)}{4}T_{ab}.$$

The nonnegative variance of every real linear combination of the J_a implies $\mathbb{1} + (N-1)T - N\mathbf{r}\mathbf{r}^{\mathsf{T}} \succeq 0$. Testing (11) on real vectors gives $A \succeq 0$. Consequently,

$$0 \preceq A \preceq \frac{N}{N-1}\mathbb{1}_3 \preceq \frac{3}{2}\mathbb{1}_3, \qquad \operatorname{Tr} A = 2. \tag{12}$$

Only the uniform bound $A \preceq 3\mathbb{1}_3/2$ is needed below.

Write $\rho' = \Phi(\rho)$. The partial trace intertwines rotations, so the rank-one and rank-two parts of the reduction acquire the same multipliers $f_1 = f$ and $f_2 = g$ as the global tensor components. Equivalently, the above first and second moments give directly

$$\mathbf{r}' = f\mathbf{r}, \quad T' = gT + \frac{1-g}{3}\mathbb{1}_3, \quad A' = \alpha\mathbb{1}_3 + gA, \qquad \alpha = \frac{2(1-g)}{3}. \tag{13}$$

4 Determinant monotonicity

We prove two elementary estimates for A and A'.

Lemma 2. *Under (12) and (3),* $\det A' \geq \det A$.

Proof. Write the eigenvalues of A as $2/3 + x_i$, where $\sum_i x_i = 0$, and set $s_2 = \sum_i x_i^2$, $p = x_1x_2x_3$. Expansion gives

$$\det A' - \det A = \frac{1-g^2}{3}s_2 - (1-g^3)p. \tag{14}$$

We claim $p \leq 5s_2/36$. This is immediate if $p \leq 0$. If $p > 0$, exactly one x_i is positive, so after relabeling $(x_1, x_2, x_3) = (q + r, -q, -r)$ with $q, r > 0$. The upper eigenvalue bound gives $q + r \leq 5/6$, whereas $s_2 = 2(q^2 + qr + r^2) \geq 6qr$. Hence $p = (q + r)qr \leq 5s_2/36$, as claimed. Since $g \leq 1$, (14) now gives

$$\det A' - \det A \geq \frac{(1-g)s_2}{36}(7 + 7g - 5g^2) \geq 0,$$

because $7 + 7g - 5g^2 = 1 + (5g + 3)(2 - g) \geq 1$ on $[-3/5, 1]$. □

Lemma 3 (Cofactor comparison)**.** *If b, c are any two eigenvalues of A, and $b' = \alpha + gb$, $c' = \alpha + gc$, then*

$$b'c' \geq f^2 bc. \tag{15}$$

Proof. We have $0 \leq b, c \leq 3/2$ and $b + c \leq 2$. Also $\alpha + gx \geq 0$ for $0 \leq x \leq 3/2$: for $g \geq 0$ this is immediate, and for $g < 0$ it follows from

$$\alpha + gx \geq \alpha + \frac{3}{2}g = \frac{4 + 5g}{6} \geq \frac{1}{6}. \tag{16}$$

Write $s = b + c$, $p = bc$.

If $g \geq -1/2$, then $2g + \alpha = 2(1 + 2g)/3 \geq 0$. The identity

$$4\big[b'c' - (g + \alpha)^2 bc\big] = \alpha\big[(2g + \alpha)(b - c)^2 + (2 - s)\{2\alpha + (2g + \alpha)s\}\big] \geq 0$$

and $(g + \alpha)^2 = (2 + g)^2/9 \geq f^2$ prove the claim.

If $-3/5 \leq g < -1/2$, Lemma 1 gives $f^2 < 1/9$. Set $h = g^2 - 1/9 > 0$. The inequality $(3/2 - b)(3/2 - c) \geq 0$ gives $p \geq 3s/2 - 9/4$. Furthermore

$$\alpha g + \frac{3}{2}h = \frac{(5g - 1)(g + 1)}{6} < 0.$$

It follows that

$$\begin{aligned} b'c' - \frac{bc}{9} &= \alpha^2 + \alpha g s + h p \\ &\geq \alpha^2 - \frac{9}{4}h + \left(\alpha g + \frac{3}{2}h\right) s \\ &\geq \alpha^2 + 2\alpha g + \frac{3}{4}h = \frac{13 + 16g - 5g^2}{36} \geq \frac{2}{45} > 0. \end{aligned}$$

The last bound follows from $13 + 16g - 5g^2 = 8/5 + (g + 3/5)(19 - 5g)$. This proves (15) in both cases. □

We can now show

$$\det \rho_2' \geq \det \rho_2. \tag{17}$$

In a common eigenbasis of A, A', with respective eigenvalues a, b, c and a', b', c', all six eigenvalues are nonnegative. Set $\Delta = 8 \det \rho_2$, $\Delta' = 8 \det \rho_2'$. Equation (11) gives the polynomial identity

$$\begin{aligned} abc(\Delta' - \Delta) = (a'b'c' - abc)\Delta \\ + aa' r_1^2 (b'c' - f^2 bc) + bb' r_2^2 (a'c' - f^2 ac) + cc' r_3^2 (a'b' - f^2 ab) \geq 0. \end{aligned}$$

Every term is nonnegative by Lemmas 2 and 3, and $\Delta \geq 0$ follows from $\rho_2 \succeq 0$. If $abc > 0$, division proves (17). If A is singular, then (11), $A \succeq 0$ and $\det \rho_2 \geq 0$ force $\det \rho_2 = 0$. The inequality is then immediate from $\rho_2' \succeq 0$. This also covers all boundary states.

5 From determinant and purity to fidelity

Orthogonally decompose the reduction into its tensor ranks:

$$\rho_2 = \frac{\mathbb{1}_3}{3} + X_1 + X_2, \qquad \rho_2' = \frac{\mathbb{1}_3}{3} + f X_1 + g X_2.$$

The Hermitian X_1, X_2 are traceless and Hilbert–Schmidt orthogonal. The bounds $|f|, |g| \leq 1$ therefore imply

$$\operatorname{Tr}[(\rho_2')^2] = \frac{1}{3} + f^2 \operatorname{Tr}(X_1^2) + g^2 \operatorname{Tr}(X_2^2) \leq \operatorname{Tr}(\rho_2^2). \tag{18}$$

For a positive trace-one 3×3 matrix B, let $e_2(B) = (1 - \operatorname{Tr} B^2)/2$ and $e_3(B) = \det B$. Equations (17) and (18) show that both e_2 and e_3 increase. The following elementary fact completes the argument without an integral representation of the square root.

Lemma 4 (Three-eigenvalue comparison)**.** *If $B, C \succeq 0$ are 3×3 matrices of trace one with $e_2(C) \geq e_2(B)$ and $\det C \geq \det B$, then $\operatorname{Tr} \sqrt{C} \geq \operatorname{Tr} \sqrt{B}$.*

Proof. Let $x, y, z \geq 0$ be the square roots of the eigenvalues of B. Write $v = x + y + z$, $d = xyz = \sqrt{\det B}$ and $e = e_2(B)$. Since $x^2 + y^2 + z^2 = 1$, elementary expansion gives

$$v \geq 1, \qquad (v^2 - 1)^2 = 4e + 8dv, \qquad v(v^2 - 1) \geq 18d. \tag{19}$$

For the last inequality, use $v^2 - 1 = 2(xy + xz + yz)$ and

$$v(xy + xz + yz) - 9xyz = z(x - y)^2 + x(y - z)^2 + y(z - x)^2 \geq 0.$$

Define u, d', e' in the same way for C. If $u < v$, then $u \geq 1$, $d' \geq d$ and $e' \geq e$ imply

$$(v^2 - 1)^2 - (u^2 - 1)^2 = 4(e - e') + 8(dv - d'u) \leq 8d(v - u).$$

Dividing by $v-u>0$ gives $(v+u)(v^2+u^2-2)\le 8d$. But

$$(v+u)(v^2+u^2-2)=v(v^2-1)+u(v^2-1)+(v+u)(u^2-1)>v(v^2-1)\ge 18d\ge 8d,$$

a contradiction. Here strictness follows from $u\ge 1$ and $v>u$. □

Apply Lemma 4 to $B=\rho_2$, $C=\rho_2'$. Then $\operatorname{Tr}\sqrt{\rho_2'}\ge\operatorname{Tr}\sqrt{\rho_2}\ge 0$; squaring and using (1) proves (2). This proves precisely the fidelity part of Conjecture 1 at $t=2$. □

6 Formal verification and reproducibility

The accompanying Lean 4.24.0 files certify the complete scalar algebraic reduction of the argument, including every $N\ge 3$, arbitrary probability mixtures, and singular inputs. The main theorem is `SU2T2.channel_fidelity_monotone_from_scalar_reduction`. Its explicit hypotheses encode the multiplier formulas, the marginal constraints, and the spectral purity and determinant identities. The files contain no unfinished proofs or additional axioms. Their exact scope, dependency pins, checked-source hashes, and build instructions are recorded in `lean/README.md`.

This is not an end-to-end Lean formalization of Theorem 1: the Choi/Casimir reduction, the physical marginal constraints, and the identification with spectral data are proved above but are not themselves formalized in Lean. This distinction is part of the certificate's scope.

Remark 1 (Why the marginal constraint matters)**.** *The bound $A\preceq 3\mathbb{1}_3/2$ comes from the symmetric extension to $N\ge 3$ qubits. An arbitrary spin-one density matrix need not obey it. The proof therefore does not assume that the induced reduced map is a positive or completely positive map on the whole spin-one state space.*